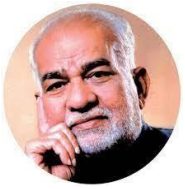
BVRIT HYDERABAD
College of Engineering for Women

(UGC Autonomous)

Department of Information Technology



17 URBAN ATLAS: SATELLITE IMAGERY FOR URBAN ANALYSIS

ABSTRACT

Urban Atlas is a deep learning-based system that segments satellite images to identify land-use regions like buildings, roads, water, vegetation, and barren land. It uses a YOLO26 model for pixel-level classification and generates segmentation masks. The system provides land distribution insights through charts and evaluates performance using precision, recall, and mAP, displaying results on an interactive dashboard.

UNIQUENESS

• Pixel-level Segmentation	• Real Time Analysis
• Interactive Dashboard	• Area Distribution Insights

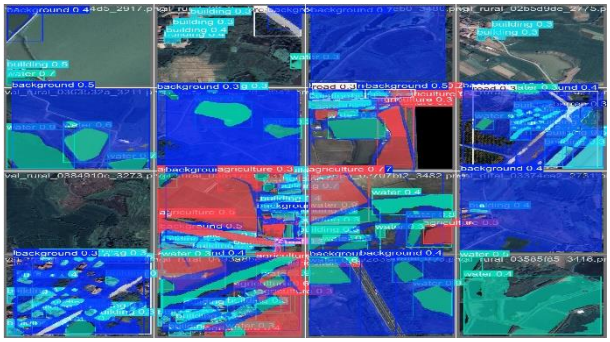
METHODOLOGY

The methodology uses the YOLO26 segmentation model for land-use classification. Satellite images are processed and passed through convolutional layers for feature extraction. The model performs pixel-level classification to generate segmentation masks for different land categories. Iterative training improves accuracy and performance. The system is deployed using Streamlit, enabling real-time analysis and visualisation through an interactive web interface.

SOCIETAL USE

The system supports SDG-11 (Sustainable Cities and Communities) by enabling efficient urban planning and environmental monitoring through satellite image analysis. It helps in understanding land-use patterns, managing resources, and making better decisions for sustainable development and smart city planning.

Land-use Segmentation



Results & Analysis

REFERENCES

[1] I. Dimitrovski, V. Spasev, S. Loshkovska, and I. Kitanovski, “U-net ensemble for enhanced semantic segmentation in remote sensing imagery,” Remote Sensing, vol. 16, no. 12, p. 2077, 2024.

[2] Ultralytics, “Yolo26 model documentation,” 2024

GITHUB LINK

https://github.com/roshika1235/UrbanAtlas-Satellitle-imagery-dataset-for-urban-analysis

22WH1A1229	22WH1A1231	22WH1A1235	22WH1A1237
Vijaya Lakshmi	K.Mamatha	Ch.Roshika	Sk.Tehazeeb

Faculty Mentor : N Sandhya Email Id:sandhya.n@bvrithyderabad.edu.in